WORKED SOLUTION

STEP II 1995, Question 7

Pure • Heron's Formula

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s2-q7

The key idea

The incircle is the hero throughout. Its radii to the points of contact are perpendicular to the sides, so they are ready-made heights for the three small triangles that tile ABC, and that gives (i). Part (ii) is the classical derivation of Heron's formula: write $\sin^2\alpha = 1 - \cos^2\alpha$, use the cosine rule to remove the angle, and then factorise by the difference of two squares, twice. For the sphere, look at a vertical cross-section. The sphere's contact points with the edges of the hole turn out to be exactly the points where the incircle touches the sides.

Solution

Part (i)

Let the incircle touch BC, CA and AB at P, Q and R'. A radius to a point of tangency is perpendicular to the tangent, so $IP \perp BC$, $IQ \perp CA$ and $IR' \perp AB$, and each of these has length r. So triangle BIC has base a and perpendicular height r, and similarly for the other two:

$$[BIC] = \tfrac{1}{2}ar, \qquad [CIA] = \tfrac{1}{2}br, \qquad [AIB] = \tfrac{1}{2}cr.$$

Since I lies inside the triangle, these three triangles fit together exactly to make ABC, with no overlaps. Adding,

$$\Delta = \tfrac{1}{2}r(a+b+c) = rs,$$

as required.

Part (ii)

Squaring $\Delta = \frac{1}{2}bc\sin\alpha$ and using $\sin^2\alpha = 1 - \cos^2\alpha$,

$$\Delta^2 = \tfrac{1}{4}b^2c^2(1-\cos^2\alpha) = \tfrac{1}{16}[4b^2c^2 - (2bc\cos\alpha)^2],$$

which is the first result. The bracket is a difference of two squares:

$$16\Delta^2 = (2bc - 2bc\cos\alpha)(2bc + 2bc\cos\alpha).$$

By the cosine rule, $2bc\cos\alpha = b^2 + c^2 - a^2$. Substituting this into each factor,

$$2bc - 2bc\cos\alpha = a^2 - (b^2 - 2bc + c^2) = a^2 - (b-c)^2,$$
$$2bc + 2bc\cos\alpha = (b^2 + 2bc + c^2) - a^2 = (b+c)^2 - a^2,$$

so

$$\Delta^2 = \tfrac{1}{16}[a^2 - (b-c)^2][(b+c)^2 - a^2],$$

the second result. Each bracket is again a difference of two squares:

$$16\Delta^2 = (a-b+c)(a+b-c)(b+c-a)(b+c+a).$$

Now we bring in s. Since $a+b+c = 2s$,

$$b+c-a = 2(s-a), \qquad a-b+c = 2(s-b), \qquad a+b-c = 2(s-c),$$

so the product is $2s\cdot 2(s-a)\cdot 2(s-b)\cdot 2(s-c) = 16\,s(s-a)(s-b)(s-c)$. Therefore $\Delta^2 = s(s-a)(s-b)(s-c)$. Since $\Delta > 0$, we take the positive square root:

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)}.$$

Part (iii)

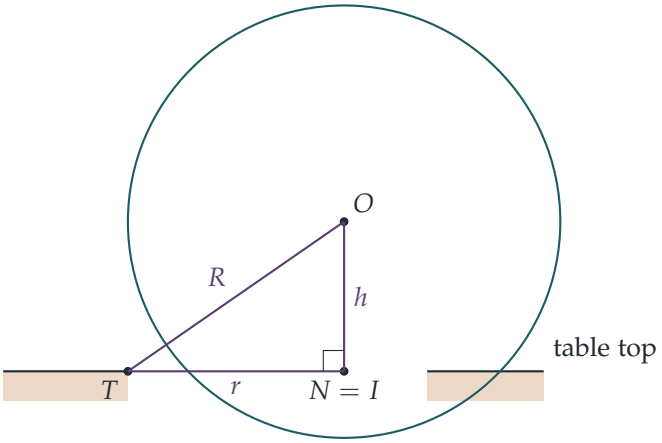
Let O be the centre of the sphere and h its height above the table top. Let N be the foot of the perpendicular from O to the plane of the table, so $ON = h$. The sphere rests on the three edges of the hole. Suppose it touches the edge BC at the point T.

Ask yourself Why must N be the incentre I?

At a point where a sphere touches a line, the radius is perpendicular to the line, so $OT \perp BC$ and $OT = R$. Also ON is perpendicular to the whole table plane, so $ON \perp BC$. So BC is perpendicular to two non-parallel lines in the plane ONT, which makes it perpendicular to that plane, and in particular to NT. So NT is the perpendicular distance from N to BC, and by Pythagoras in the right-angled triangle ONT,

$$NT^2 = R^2 - h^2.$$

The same argument for the edges CA and AB shows that N is at the same distance $\sqrt{R^2 - h^2}$ from all three sides. The point N lies inside the triangle (the sphere sits in the hole, above it), and the only point inside a triangle that is equidistant from all three sides is the incentre. So $N = I$, the common distance is r, and the points of contact are exactly where the incircle touches the sides. These lie on the edges themselves, not on their extensions, so the sphere really does rest on the three edges.



The diagram shows the vertical section through O and T; the edge BC passes through T, perpendicular to the page. Pythagoras in triangle ONT now gives $h^2 = R^2 - r^2$. From (i) and (ii),

$$r = \frac{\Delta}{s} = \frac{\sqrt{s(s-a)(s-b)(s-c)}}{s}, \qquad \text{so} \qquad r^2 = \frac{(s-a)(s-b)(s-c)}{s}.$$

Hence the centre of the sphere is at height

$$h = \sqrt{R^2 - \frac{(s-a)(s-b)(s-c)}{s}}$$

above the table top. This needs $R > r$: a sphere no bigger than the incircle falls straight through the hole.

Checks

Take the 3–4–5 triangle: $s = 6$, and Heron gives $\Delta = \sqrt{6 \cdot 3 \cdot 2 \cdot 1} = 6$, which matches $\frac{1}{2} \cdot 3 \cdot 4$. Then $r = \Delta/s = 1$, which you can confirm from the right-angle formula $r = \frac{1}{2}(3 + 4 - 5)$. With $R = 2$ the formula gives $h = \sqrt{4 - 1} = \sqrt{3} \approx 1.732$. As $R \to r^+$, $h \to 0$: the sphere sinks until its equator is level with the table and sits on the incircle. As a broader check, we tested 18,816 random triangles numerically. In every case, a centre placed at height h above the incentre was at distance exactly R from all three edge lines, with the nearest points inside the edges.

Common slips

- Taking $h = \sqrt{R^2 - d^2}$, where d is the distance from I to a *vertex*. The sphere touches the edges, not the corners.
- Assuming without reason that the centre is above the incentre. It needs the perpendicularity argument, or at least a clear symmetry reason.
- Mixing up the factor pairs in (ii): $b + c - a = 2(s - a)$, not $2(s - b)$ or $2(s - c)$.

Beyond STEP

Heron's formula belongs to a family of symmetric formulae for areas and volumes. Brahmagupta's formula does the same job for a cyclic quadrilateral, and the Cayley–Menger determinant gives the volume of a tetrahedron from its six edge lengths. The step "radius $\perp$ tangent line, so the foot of the perpendicular is the incentre" is the three-dimensional version of the theorem of the three perpendiculars, which is worth knowing for vector geometry. For many more questions of this kind, see Siklos, *Advanced Problems in Mathematics*.

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